

Literature answer for arXiv:0909.0131

Every bounded truncated Toeplitz operator has a Carleson-measure quasisymbol

Status. `literature__already__answered`. This is an exact later theorem, not a new proof produced by the run.

Source question. Remark 5.3 of Baranov–Chalendar–Fricain–Mashreghi–Timotin [1], source PDF page 19, asks whether every bounded truncated Toeplitz operator in $\mathcal{T}(K_\Theta)$ is of the form A_μ^Θ for some complex Carleson measure μ for K_Θ . Here this means that $|\mu|$ induces a bounded embedding $K_\Theta \hookrightarrow L^2(|\mu|)$ and

$$\langle A_\mu^\Theta f, g \rangle = \int_{\mathbb{T}} f \bar{g} d\mu.$$

Supporting answer. Baranov–Bessonov–Kapustin [2] define $\mathcal{C}_2(\theta)$ as the finite complex Borel measures μ for which $K_\theta \hookrightarrow L^2(|\mu|)$ is bounded, and call such a measure a quasisymbol when the displayed identity holds. Their introduction explicitly identifies as Sarason’s conjecture the assertion that every bounded truncated Toeplitz operator has such a quasisymbol and says that they prove it. Theorem 2.1(2), supporting PDF page 6, states:

Any bounded truncated Toeplitz operator on K_θ admits a quasisymbol from $\mathcal{C}_2(\theta)$.

This is exactly the affirmative answer requested in the source.

Scope. The result answers the Carleson-measure representation question only. It does not settle the source’s general reproducing-kernel thesis (Question 3) or the later conjecture characterizing the inner functions for which every bounded truncated Toeplitz operator has a bounded L^∞ symbol.

Search evidence. Cheap run indexes and bounded web/arXiv searches through 2026-08-09 used arXiv:0909.0131, the exact question, “quasisymbol”, “Carleson measure”, and truncated Toeplitz symbols. They located the explicit later theorem arXiv:1009.5123 and corroborating surveys.

References

- [1] A. Baranov, I. Chalendar, E. Fricain, J. Mashreghi, and D. Timotin, *Bounded symbols and reproducing kernel thesis for truncated Toeplitz operators*, J. Funct. Anal. 259 (2010), 2673–2701; arXiv:0909.0131.
- [2] A. Baranov, R. Bessonov, and V. Kapustin, *Symbols of truncated Toeplitz operators*, J. Funct. Anal. 261 (2011), 3437–3456; arXiv:1009.5123, doi:10.1016/j.jfa.2011.08.005.